Translating Modern English into Old English Using Transformer-Based Language Models

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Abstract

This project proposes a neural machine translation system for translating **Modern English into Old English** using transformer-based language models. Although neural translation performs well for many modern languages, translating into historical language variants such as Old English is challenging due to large linguistic differences and the lack of parallel data.

We plan to build a transformer-based model initialized with a pretrained Modern English encoder and extend it with additional transformer layers to generate Old English text. The training data will consist of paired Modern–Old English sentences generated and filtered using large language models. The system will be evaluated using standard machine translation metrics along with a qualitative analysis of translation quality. Through this project, our aim is to study how synthetic data and transformer models can be applied to low-resource historical language translation tasks.

1 Introduction

Translating Modern English into Old English is a challenging task due to substantial differences in vocabulary, grammar, spelling, and sentence structure. Unlike modern English variants, Old English relies heavily on morphological inflection and allows more flexible word order, which makes direct translation difficult even for fluent modern English speakers. As a result, automated translation tools for Old English remain limited.

From an NLP perspective, this task is non-trivial because it goes beyond simple word-level substitution and requires modeling significant structural and morphological differences between the two language variants. In addition, large-scale aligned Modern–Old English datasets are scarce, which makes standard supervised machine translation approaches difficult to apply directly. These factors together make Modern-to-Old English translation a complex low-resource sequence modeling problem.

2 Summary

(a) Proposed Software: The goal of this project is to implement an end-to-end neural machine translation system that translates Modern English sentences into Old English. The system will be built using a transformer-based sequence-to-sequence architecture and trained in a supervised learning setting.

The model will take tokenized Modern English text as input and generate Old English translations as output. We plan to initialize the encoder with a pretrained Modern English language model and extend the architecture with additional transformer layers to better model the syntactic and morphological properties of Old English. The decoder will be trained to generate Old English tokens autoregressively based on the encoded input representations. We will try to adapt a tokenization scheme based on Modern English for Old English. The model should be trained and design in such a way that while supervised training mostly consists of translating from Modern English to Old English, the model should theoretically be able to translate from Old English to Modern English with ease. We will also try to experiment with unsupervised learning methods of designing an autoencoder that works directly on Old English Corpus.

(b) Data:

We have listed some data we plan to use and experiment during the project, since it is in a proposal stage, we cannot guarantee if we will use all the data, or if additional datasets are later incorporated. We will use these corpus to first to unsupervised learning to form embeddings and representations, then use small LLMs to translate Old English to Modern English or the other way around for us to do supervised finetuning that actually does translation.

Old English / Modern-to-Old-English Relevant Corpora

- Dictionary of Old English Corpus (DOEC)
- York–Toronto–Helsinki Parsed Corpus of Old English Prose (YCOE)
- Helsinki Corpus (Old English section)
- Consolidated Library of Anglo-Saxon Poetry (CLASP)
- Old English Poetry Project (Rutgers)
- Project Gutenberg (Old English texts and translations)
- Tatoeba (sentence translation pairs)
- OPUS (parallel corpora collection)

Modern English Corpora (for pretraining / auxiliary data)

- Common Crawl
- C4 (Colossal Clean Crawled Corpus)
- English Wikipedia Dumps
- British National Corpus (BNC)
- Corpus of Contemporary American English (COCA)
- Project Gutenberg (English books)
- OpenWebText
- The Pile
- WikiText (e.g., WikiText-103)

(c) Evaluation:

For evaluation, we will measure translation quality on a held-out test set using both automatic metrics and qualitative analysis. First, we will report *SacreBLEU*, *chrF++*, and *TER* to quantify lexical overlap, character-level similarity, and edit distance between model outputs and reference Old English translations.

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Second, we will run human evaluation on a sampled subset, where we will have modern english speakers score outputs for (1) semantic adequacy, (2) grammar completeness, and (3) Old English stylistic plausibility.

Third, we will compare our method against strong baselines, including a rule/dictionary-based system, a standard transformer without pretrained encoder initialization, as well as a small LLM as a upper limit.

We will also evaluate the computational efficiency of our model, if it can run and execute on low-end hardware and the possibility for real-time inference, which can benefit users who are reading old-english text.

(d) Timeline and Team Responsibilities. The project will be completed in several phases over the semester:

- **Late February:** finalize project scope, set up the codebase and experiment pipeline, and prepare initial data collection/generation scripts.
- **Early March:** generate synthetic Modern–Old English sentence pairs, collect seed corpus data, and begin data cleaning and normalization.
- **Mid March:** complete preprocessing and tokenization, create train/dev/test splits, and implement baseline translation systems.
- **Late March:** implement the proposed pretrained-encoder transformer model and run initial training experiments.
- **Early April:** perform hyperparameter tuning, compare against baselines, and refine data filtering strategies.
- **Mid April:** run final evaluation (automatic metrics + qualitative analysis), conduct ablation studies, and complete error analysis.
- **Late April:** finalize the report and presentation materials, consolidate results, and rehearse the final demo/presentation.

Yidong Zhang will focus on overall system design and model architecture, including defining the transformer-based translation framework.

Yuang Cai will be responsible for generating synthetic Modern–Old English sentence pairs using large language models and refining the data generation process.

Jingcheng Yang will handle data preprocessing tasks such as text normalization, tokenization, and dataset preparation.

Duy Anh Hong will assist with model training, hyperparameter tuning, and evaluation of translation performance. All team members will contribute to result analysis and the preparation of the final report.

3 Initial Evaluation

To obtain an initial estimate of task difficulty, we evaluated a simple retrieval-based baseline on the validation set. In this baseline, each Modern English source sentence in the training set is represented using TF-IDF features with unigram and bigram statistics. For each validation input, we compute cosine similarity against all training source sentences, retrieve the nearest neighbor, and use its paired Old English sentence as the predicted translation.

Using this approach, our retrieval baseline achieved a corpus-level BLEU score of 12.54 on the validation set. This result suggests that simple lexical matching is able to recover partially relevant translations in some cases, but overall performance remains limited. In particular, the baseline performs best when the validation sentence is very close to a sentence already seen in the training set, such as near-duplicates or paraphrases with only minor surface-level differences.

0	he was at the gate of death.	hē wæs æt þæm geate deaðes.	he was a good cat.	hē wæs gōd catt.	0.437722
1	let's eat grandma.	utan etæðeoðer ælan.	let's begin.	onginnaþ.	0.446549
2	but that your fear of me was greater than his. ...	'ac þe wæs ælc ege māre þonne hīs. ÞÆ, þe ic n...	that's your book.	þæt is þīn bōc.	0.270131
3	my dad went to buy milk.	min fæder wōde mealca brycgian.	buy!	bygæ!	0.221126
4	it is john's apple.	þæ is Iohannes æppel.	my name is john.	mīn nama hīg Iohannes.	0.459984
5	our father who art in heaven, hallowed be your...	fæder ure þū þe eart on heofonæ, sī þīn nama ...	our father who art in heaven, hallowed be thy ...	fæder ure þū þe eart on heofonæ, sī þīn nama ...	0.409192
6	the child ran into the house.	þæt cild arn on þæt hūs.	the dog is in the house.	se hund is on þæm hūse.	0.420888
7	just when he was beginning the task, the angel...	nū þæm þe he wolde þæt weorc beginnan, þa cil...	he was a good cat.	hē wæs gōd catt.	0.240589
8	i don't understand.	ic ne forstande.	i do not understand.	ic ne forstande.	0.405014
9	you are everything to me.	þū hīst eall tō mē.	you're everything to me.	þū hīst eall tō mē.	0.405532

BLEU score: 12.5449030050177

Figure 1: Examples of retrieval-based predictions on the validation set.

Qualitative inspection further reveals the strengths and weaknesses of this method. For example, the baseline works well on inputs whose wording strongly overlaps with a training example, since the retrieved sentence often has a correct or nearly correct Old English counterpart. However, performance degrades substantially when the input contains unseen phrasing, lower lexical overlap, or compositional meaning that cannot be captured through nearest-neighbor matching alone. In such cases, the retrieved sentence may be only superficially similar and the resulting Old English output can be semantically unrelated to the reference translation.

Overall, these initial results show that retrieval provides a reasonable lower-bound benchmark for the task, but it is insufficient as a complete translation solution. In the next stage of the project, we plan to compare this baseline with a learned sequence-to-sequence model that can better generalize beyond sentence-level memorization and surface-form similarity.

4 Insights

Our progress so far provides several useful insights about both the dataset and the difficulty of the Modern English to Old English translation task.

First, the dataset is relatively small, containing only 404 sentence pairs. Although the data is sufficient for initial experimentation and baseline development, its limited size makes the task challenging for any model that must generalize beyond memorized patterns. In addition, the English and Old English sides have comparable average sentence lengths (6.13 and 5.85 tokens, respectively), which suggests that the translation task is mostly sentence-level rather than involving large structural expansions. However, the Old English vocabulary size (1121) is larger than the Modern English vocabulary size (920), indicating greater lexical variation and potentially richer inflectional or orthographic diversity on the target side.

Second, the retrieval baseline shows that simple lexical overlap can partially solve the task, but only in restricted cases. The model performs reasonably well on sentences that are near-duplicates or close paraphrases of examples already seen in the training set, such as minor wording changes or contractions. However, it performs poorly when the input differs more substantially from the training data, even if some keywords overlap. For instance, sentences such as "let's eat grandma." and "my dad went to buy milk." retrieve superficially related but semantically incorrect examples, leading to clearly wrong translations. This indicates that TF-IDF retrieval captures shallow word-level similarity rather than true semantic equivalence.

Third, these early results suggest that the main challenge of the project is not only lexical mapping, but also semantic preservation and generalization. A baseline based on nearest-neighbor retrieval cannot generate new translations, handle unseen constructions, or model compositional meaning. Therefore, while retrieval is a useful lower-bound benchmark, stronger sequence-to-sequence or transformer-based methods will likely be necessary to produce more reliable translations.

Overall, our findings so far suggest that the dataset is adequate for a milestone-stage prototype, but the task remains difficult because of limited data size, target-side lexical diversity, and the inability of simple baselines to move beyond surface-form matching.

5 Model Description

5.1 Architecture Overview

We implement a hybrid encoder-decoder architecture for Modern English to Old English translation. The pipeline follows:

Modern English → BERT Encoder → Linear Projection → Transformer Decoder → Old English.

5.2 Encoder

We use `bert-base-uncased` as the encoder, a 12-layer bidirectional transformer pretrained on large English corpora. Given a tokenized Modern English sentence, BERT produces contextual embeddings of shape (batch, seq_len, 768). BERT parameters are frozen during training to prevent overfitting on our small dataset and to preserve pretrained representations.

A linear projection layer maps the 768-dimensional BERT output to the decoder's hidden dimension of 256:

$$\mathbf{h}_{\text{enc}} = \mathbf{W}_{\text{proj}} \cdot \text{BERT}(x) \in \mathbb{R}^{\text{seq_len}\times 256}$$

5.3 Decoder

The decoder is a 2-layer Transformer decoder with the following components per layer:

- **Masked Self-Attention:** prevents the decoder from attending to future tokens during training, ensuring autoregressive generation.
- **Cross-Attention:** allows each decoder position to attend over the full encoder output, enabling the model to align Old English tokens with relevant Modern English context.
- **Feed-Forward Network:** two linear layers (256 → 512 → 256) with ReLU activation.
- **Layer Normalization and Dropout** (rate 0.1) are applied after each sub-layer.

Sinusoidal positional encodings are added to the target embeddings before the first decoder layer to inject positional information.

5.4 Target Tokenizer

We build a custom word-level tokenizer for Old English trained on the training set vocabulary. Special tokens <bos>, <eos>, <pad>, and <unk> are included. The final vocabulary size is approximately 1,400 tokens.

5.5 Training

The model is trained with teacher forcing: the decoder input is the target sequence shifted right (prepended with <bos>), and the training objective is cross-entropy loss with label smoothing of 0.1, ignoring padding positions:

$$\mathcal{L} = -\sum_{t=1}^{T} \log P(y_t \mid y_{<t}, \mathbf{h}_{\text{enc}})$$

We use the Adam optimizer with learning rate 10^{-3} and gradient clipping at 1.0. Training uses early stopping with patience of 10 epochs, saving the checkpoint with the lowest validation loss.

5.6 Data

We collected 1,191 Modern English–Old English sentence pairs from Tatoeba, a crowd-sourced multilingual sentence database. The dataset is split 80/10/10 into train (952), validation (119), and test (120) sets.

6 Final Evaluation

6.1 Automatic Metrics

We evaluate our model on the held-out test set using corpus-level BLEU score computed via SacreBLEU. Table 1 summarizes the results.

Table 1: BLEU scores on the test set.

Model	BLEU
TF-IDF Retrieval Baseline	12.54
BERT + Transformer Decoder (ours)	**63.48**

Our model achieves a BLEU score of 63.48, representing a substantial improvement of over 50 points compared to the retrieval baseline. This result demonstrates that a learned sequence-to-sequence model can generalize far beyond surface-level lexical matching.

6.2 Qualitative Analysis

Table 2 shows representative translation examples from the test set.

Table 2: Example translations on the test set.

Modern English	Predicted Old English
god is everywhere.	god biþ æġhwǣr.
tom is dead.	tom is dēad.
i do not know him.	ič ne cann hine.
the king went to the hall.	se cyning cōm.

The model handles short, common sentences well, often producing exact or near-exact matches to the reference. Performance degrades on longer sentences and on vocabulary absent from the training set, which is expected for a low-resource setting.

7 Discussion

7.1 Key Findings

The transformer-based model substantially outperforms the retrieval baseline, confirming that learned sequence-to-sequence models can better capture the structural and morphological transformations required for Modern-to-Old English translation. Expanding the dataset from 404 to 1,191 sentence pairs through automated collection from Tatoeba was the single most impactful improvement, reducing validation loss from 5.74 to 2.46 and raising BLEU from approximately 19 to 63.48.

7.2 Limitations

Despite strong performance on the test set, the model has several limitations. First, the dataset remains small by NMT standards, and the model struggles with out-of-vocabulary words. Second, freezing BERT limits the encoder's ability to adapt to translation-specific features. Third, greedy decoding is used at inference time, which can produce repetitive outputs for certain inputs.

7.3 Future Work

Future directions include fine-tuning BERT on Old English data, incorporating beam search decoding, and expanding the training corpus with additional Old English sources such as the Dictionary of Old English Corpus (DOEC) or Project Gutenberg bilingual texts.

8 Reference

https://github.com/yuangc3/CS410_Old_English_Tranlation